

Advanced (Habanero)

- Q1.** What is the primary purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the median and quartiles of a dataset
(C) To illustrate the mode of a dataset
(D) To compare two datasets
(E) To calculate the standard deviation
- Q2.** Which part of the box-and-whisker plot represents the interquartile range (*IQR*)?
(A) The box
(B) The whiskers
(C) The median line
(D) The outliers
(E) The entire plot
- Q3.** A dataset has values 1, 3, 5, 7, 9. What is the Q_1 ?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9
- Q4.** What is an outlier in a dataset?
(A) A value within $1.5 \cdot IQR$ of Q_1 or Q_3
(B) A value more than $1.5 \cdot IQR$ away from Q_1 or Q_3
(C) The mean of the dataset
(D) The median of the dataset
(E) The mode of the dataset
- Q5.** Which of the following datasets is most likely to have outliers?
- (A) 1, 2, 3, 4, 5
(B) 10, 20, 30, 40, 1000
(C) 1.1, 1.2, 1.3, 1.4, 1.5
(D) 10, 10, 10, 10, 10
(E) 5, 5, 5, 5, 5
- Q6.** What does the length of the whisker represent in a box-and-whisker plot?
(A) The range of the dataset
(B) The interquartile range (*IQR*)
(C) The median of the dataset
(D) The minimum or maximum value
(E) The standard deviation
- Q7.** A box-and-whisker plot shows $Q_1 = 2$, $Q_3 = 6$, and the lowest value is 0. What is the *IQR*?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q8.** Which of the following statements is false about box-and-whisker plots?
(A) The plot displays the median and quartiles
(B) The plot shows the interquartile range (*IQR*)
(C) The plot can be used to compare datasets
(D) The plot always shows the mean
(E) The plot can be used to detect outliers

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset 2, 4, 6, 8, 10, 12, 14, 16, 100. Identify the quartiles and interquartile range (*IQR*).
- Q10.** Explain how to detect outliers using a box-and-whisker plot. Provide an example with a dataset.

Chef's Tips

- Tip 1: Box-and-whisker plots are useful for comparing datasets.
- Tip 2: The interquartile range (*IQR*) is a measure of dispersion.
- Tip 3: Outliers can affect the mean, but not the median.